Universidade do Minho

# Bases de Dados

Licenciatura em Ciências da Computação
Ano Letivo de 2025/2026
2º Semestre
**PL07**

**Tema:** Introdução à SQL (DDM e DML)

## Exercise 1

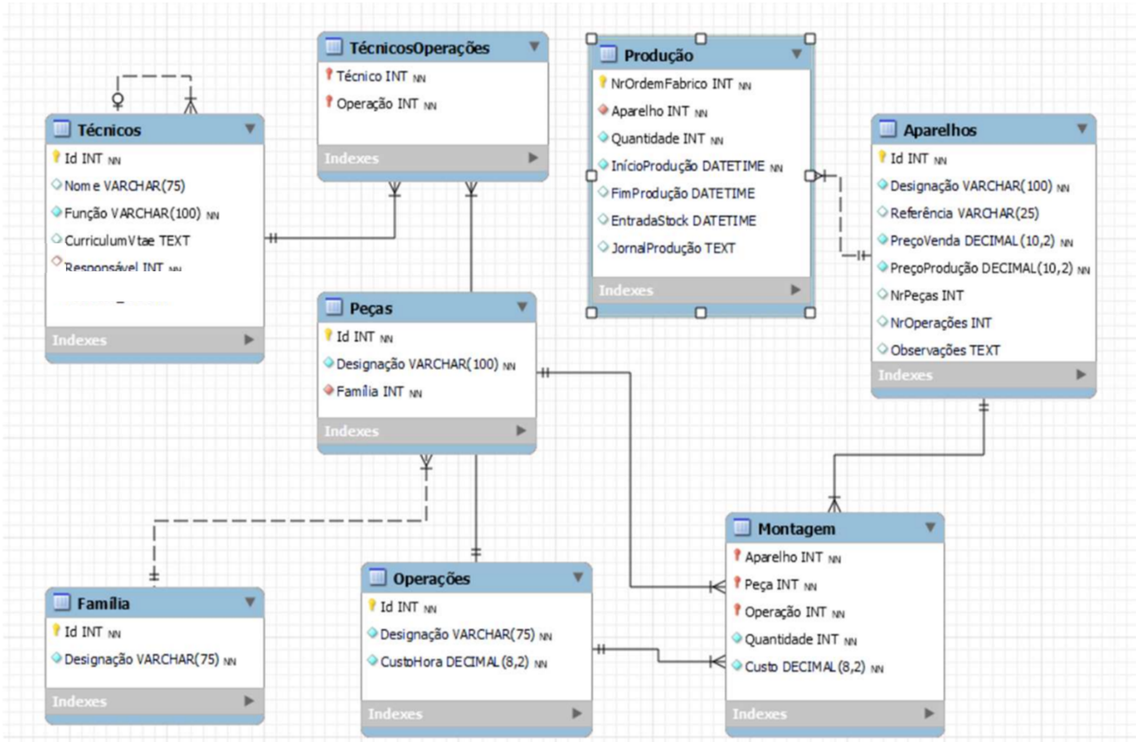


Figure 1 – Logic Schema. (Belo, O.)

Consider the logical schema presented in Figure 1, which refers to a database for a company that assembles devices. After carefully and thoroughly analyzing the given schema, answer the following SQL exercises.

1. Create a database named Fabrica, ensuring that: It is only created if it does not already exist;It is selected for use.
2. Create the necessary tables according to the provided model. When defining the tables, correctly use:

   - Appropriate data types (INT, VARCHAR, DECIMAL, DATETIME, TEXT)
   - NOT NULL constraints for mandatory attributes
   - PRIMARY KEY in all tables

Implement the following constraints:

- UNIQUE constraint on the device reference
- CHECK constraint to ensure that prices are positive
- DEFAULT value on an attribute of your choice
- FOREIGN KEY constraints to represent relationships

Define:

- A composite primary key in the TechniciansOperations table
- A composite primary key in the Assembly table

For foreign keys:

Try defining ON DELETE CASCADE in one relationship
Try defining ON UPDATE CASCADE

3. Add the field NIF to the Technicians table
4. Change the size of the technicians' Name field
5. Set the SalePrice field as NOT NULL
6. Remove an attribute of your choice from a table
7. Define a default value for an attribute
8. Remove the default value defined in the previous step
9. Add an additional UNIQUE constraint to a table
10. Insert data into the tables:
    a) Insert a record specifying all attributes
    b) Insert a record specifying only some attributes
    c) Insert multiple records in a single statement
11. Test the following behaviors:
    a) Insert a record without a value in a field that has a DEFAULT
    b) Insert a record that violates a UNIQUE constraint
    c) Insert a record that violates a FOREIGN KEY constraint
    d) Explain the results of cases b) and c)
12. Update the production price of a device to 100
13. Update multiple fields of a device (price and description)
14. Increase the hourlyCost of all operations by 10%
15. Update technicians with role = 'Junior' to 'Mid-level'
16. Update productions with quantity < 10 to 10
17. Update operations with IDs 1, 2, and 3 by increasing hourlyCost by 5
18. Execute an UPDATE without a WHERE clause. What happens?
19. Delete a record from the Operations table
20. Delete productions with quantity < 5
21. Delete all records from a table
22. Execute DELETE without a WHERE clause. What happens?
23. Try deleting a part used in Assembly. What happens?
24. Modify the foreign key to ON DELETE CASCADE and repeat
25. Try dropping a table with associated foreign keys. What happens?